

Ruichen Xu

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SUMMARY

Ph.D. candidate in Computational Applied Mathematics building **Python/PyTorch models and reproducible training and evaluation pipelines**. Research spans sparse-observation inference, scientific representation learning, generative models and optimization, with a **560,000-record benchmark** and coauthored ICLR, TMLR and IJCNN papers. Expected graduation: 2027.

EDUCATION

Stony Brook University Ph.D., Computational Applied Mathematics GPA: 4.00/4.00 Expected 2027	Aug 2022 – Present Stony Brook, NY
New York University – Courant Institute M.S., Mathematics GPA: 4.00/4.00	Sep 2020 – May 2022 New York, NY
University of California, Davis M.S., Statistics GPA: 3.94/4.00	Sep 2019 – Jun 2020 Davis, CA
Beijing University of Chemical Technology Bachelor's degree, Financial Mathematics Minor: Commercial Management	Sep 2015 – Jun 2019 Beijing, China

TECHNICAL SKILLS

Programming & Computing: Python, PyTorch; GPU/HPC experiment workflows, SLURM.

Modeling: Representation learning, neural operators, graph learning, diffusion models, reinforcement learning.

Evaluation & Optimization: Benchmarking, ablations, distribution-shift evaluation, simulated annealing, molecular search.

RESEARCH EXPERIENCE

Stony Brook University Doctoral Research <i>Advisor: Prof. Yuefan Deng</i>	Aug 2022 – Present
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PDE-OBS: Controlled Evaluation Across Observation Patterns 2025 – Present

- Built a **560,000-record dataset** across seven PDE families to support controlled benchmarking under partial observation.
- Adapted **seven baselines** to mask-aware inputs with shared records, targets and splits to isolate observation-pattern effects.
- Evaluated **441 trained configurations** across **3,969 observation-pattern tests** on a controlled **14,000-record study subset**.
- Found higher mean cross-pattern error across evaluated PDE-method pairs, highlighting limits of matched-pattern evaluation.
- Tested mixed-pattern training in selected settings and released reproducible scripts for observation-shift evaluation. [Code]

JEPA for Science Jan 2026 – Present

- **HP-JEPA:** Developed multi-resolution joint-embedding prediction with hierarchical graph partitioning for scientific representation learning; outperformed Graph-JEPA on most evaluated downstream graph classification and regression tasks.
- **JENO:** Developed a two-stage pipeline for sparse inverse PDE recovery, combining latent pretraining using complete training fields with supervised encoder and readout fine-tuning to reconstruct physical fields from partial observations at inference.
- **SCOPE:** Coupled observation-conditioned latent prediction with a shared decoder for physical fields, supporting forward and inverse PDE inference with deterministic one-pass reconstruction from partial measurements across evaluated tasks.

ORACLE: LLM-Guided Molecular Optimization 2025 – Present

- Built a **simulated-annealing pipeline** integrating LLM-proposed molecular edits, chemical-validity checks, candidate scoring and acceptance decisions to automate iterative, multi-objective search for structure-based molecular optimization.
- Implemented configurable objectives for docking scores, drug-likeness (QED), synthetic-accessibility scores and molecular diversity to compare and rank candidate molecules and examine trade-offs between competing optimization criteria.
- Evaluated molecular edit operators, cooling schedules and acceptance rules through ablation studies, comparing optimization curves, molecular validity and Pareto trade-offs to characterize the behavior of alternative molecular search strategies.

Physics-Aware Dynamics, Diffusion & Optimization Selected 2025 Work

- **KAR-HNN / position-only dynamics:** Co-led Kolmogorov-Arnold Hamiltonian learning for structured dynamics prediction and led velocity-inferred modeling from position measurements, enabling dynamics learning without observed velocities.
- **APOD / RL-QESA:** Developed observation- and PDE-guided diffusion methods for physics-constrained sampling, alongside reinforcement-learned annealing schedules for optimization; first author of the corresponding ICML 2025 workshop papers.

Efficient Generative Models & VLM Post-Training Selected 2025–2026 Work

- **GAGA (ICLR 2026):** Coauthored research on Gaussian-aware Gaussian approximation for diffusion-based 3D molecular generation, focusing on sampling efficiency and reporting shorter generative trajectories in evaluated configurations.
- **VLM distillation (preprint):** Coauthored work on student-curriculum coupling for on-policy distillation in temporal video grounding; the method outperformed Video-OPD in mean recall while using fewer examples and less training time.

SELECTED PUBLICATIONS & MANUSCRIPTS

Name in bold; * denotes co-first author. Full list: [Homepage](#) / [Google Scholar](#)

Conference & Journal Papers

- [1] **GAGA: Gaussianity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** J. Qu, W. Gao, **Ruichen Xu**, Y. Liu. *ICLR 2026*.
- [2] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** W. Gao, **Ruichen Xu**, Y. Deng, Y. Liu. *ICLR 2025*.
- [3] **Kolmogorov-Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, **Ruichen Xu***, et al. *IJCNN 2025*. Co-first author; project lead.
- [4] **Dynamic Schwartz-Fourier Neural Operator for Enhanced Expressive Power.** W. Gao, J. Luo, **Ruichen Xu**, Y. Liu. *TMLR 2025*.
- [5] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** W. Gao, **Ruichen Xu**, H. Wang, Y. Liu. *TMLR 2024*.
- [6] **Towards Principled Test-Time Adaptation for Time Series Forecasting.** H. Wang, **Ruichen Xu**, et al. *TMLR 2026*; *accepted with minor revision*.
- [7] **Multistep Generative Backmapping of Coarse-Grained Structures.** G. Kementzidis, E. Wong, J. Nicholson, **Ruichen Xu**, Y. Deng. *Computer Physics Communications 327, 110286 (2026)*.
- [8] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** **Ruichen Xu**, C. Yu, et al. *NYSDS 2025, SIAM Proceedings*.
- [9] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** M. Cederholm, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.
- [10] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** A. Ding, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.

Workshop Papers

- [11] **APOD: Adaptive PDE-Observation Diffusion for Physics-Constrained Sampling.** **Ruichen Xu**, H. Wang, et al. *ICML 2025 Workshop on Assessing World Models (poster)*.
- [12] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** **Ruichen Xu**, K. Li, et al. *2nd AI for Math Workshop at ICML 2025 (poster)*.

Preprints & Manuscripts

- [13] **SCOPE: Observation-Conditioned Full-Target Prediction for Sparse PDE Inference.** **Ruichen Xu**, et al. *arXiv:2609.36527, 2026*; *under review at ICLR 2027*.
- [14] **PDE-OBS: Controlled Evaluation Across Observation Patterns.** **Ruichen Xu**, et al. *arXiv:2609.36521, 2026*.
- [15] **HP-JEPA: Hierarchical Partitioning for Multi-Resolution Graph Joint-Embedding Predictive Learning.** **Ruichen Xu**, et al. *arXiv:2608.00491, 2026*.
- [16] **JENO: Full-Field Latent Prediction for Sparse Inverse PDE Inference.** **Ruichen Xu**, et al. *Manuscript, 2026*.
- [17] **Less Data, Better Timing: Student-Curriculum Coupling for VLM On-Policy Distillation in Temporal Video Grounding.** J. Qiu, Y. Kim, **Ruichen Xu**, et al. *arXiv:2609.40055, 2026*.
- [18] **The Impact of Move Schemes on Simulated Annealing Performance.** **Ruichen Xu**, H. Wang, Y. Deng. *arXiv:2504.17949, 2025*.

MENTORING & TEACHING

AI3 REU, Stony Brook University | Research Mentor

Jun – Jul 2026

Mentored AI4Science project scoping, experiment design and implementation; guided reproducible workflows and research presentations.

Stony Brook University | Instructor & Teaching Assistant

Aug 2022 – Jan 2026

Instructor: Statistical Laboratory, Numerical Analysis and Differential Equations. Designed lectures and assessments; led recitations, office hours and grading as a teaching assistant.

NYU Courant | Recitation Leader & Grader

2020 – 2022

Led recitations in Mathematics for Economics II and Probability, Statistics & Decision Making; graded Analysis and Special Topics coursework.

RECOGNITION, ACADEMIC SERVICE & TALKS

ICML 2026 Gold Reviewer. Conference service: ICLR (2026/2027), NeurIPS (2026), IJCNN (2025/2026), KDD (2026/2027); AAAI 2027 Program Committee.

Journal / workshop reviewing: TMLR, IEEE TNNLS, Neurocomputing; AI for Math at ICML 2025.

IACS presentations: AI for PDEs (2025); Diffusion4PDE (2025); Active Learning for Neural Operators (2024).